

INTEGRATED AGRI EXPORT & COLD CHAIN INFRASTRUCTURE

Farm to global market, on one traceable spine.

A common-use export hub in the Mumbai / JNPA corridor, wired to a TradeTech platform that moves perishables from the field to the vessel without losing a day, a degree or a document.

6,410 MT

handled this season

257 TEU

exported, 9 countries

96%

sailed as booked

6.1%

post-harvest loss

THE PROBLEM

Indian produce does not lose value in the field. It loses value between the field and the vessel.

18–22%

of horticultural output is lost after harvest

Breaks in the cold chain, ungraded mixed lots, and produce that waits for a truck instead of a truck waiting for produce.

1 in 5

reefer bookings roll or miss the cut-off

A rolled container on a 6-day lane costs the shelf life the buyer paid for. The exporter finds out when it is already too late to re-book.

Days

lost to documentation rework per shipment

Twenty-eight controlled documents, a dozen agencies, and no single place that knows which one is missing right now.

Every one of these is a coordination failure, not an agronomy failure. Coordination is a software problem attached to a physical asset.

Five layers, one operating company

The farmer network and the port are the two ends. We own everything in between and meter it.

1 Farmer network Clusters, FPOs, collection points, contracted reefer fleet, field QC at the point of purchase.

2 Farm-to-hub logistics Graded, weighed, lot-tagged and temperature-controlled from the first hour after harvest.

3 Mumbai integrated hub Twelve zones from receiving to stuffing: lab, sorting, washing, pre-cooling, cold and CA storage, packing, palletisation.

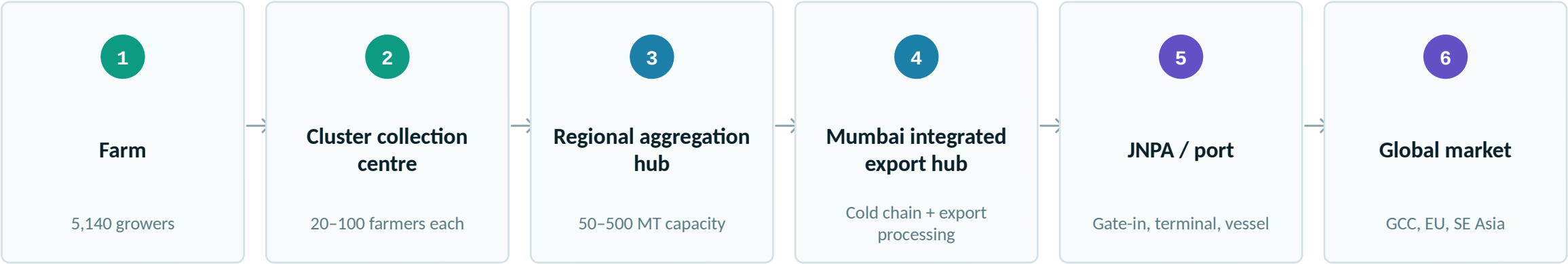
4 Export execution Export QC, documentation, container booking, customs, VGM, gate-in, vessel.

5 TradeTech platform The digital identity that follows every lot, and the control tower that protects every sailing.

Layers 1-4 are physical and fundable. Layer 5 is what makes layers 1-4 hard to copy.

Consolidate produce before it travels far

Six levels between the plant and the buyer, each one shortening the time the cargo spends uncooled.



TIME FROM HARVEST TO TEMPERATURE CONTROL

Under 6 hours, measured per lot, not per season

TRUCKS ARRIVING UNANNOUNCED AT THE HUB

Zero — every vehicle holds a booked dock slot

LOTS MIXED ACROSS GRADES

None — one lot ID, one grade, one destination

One building: packhouse, cold store, customs yard

Zones A to L. Cargo moves in one direction and is never handled twice, from receiving to container stuffing.



Cold & CA storage

900 pallet positions, commodity-specific set points

Pre-cooling

Forced-air, core temperature verified per lot

Reefer docking

Six sealed bays, stuffing under temperature

Backup power

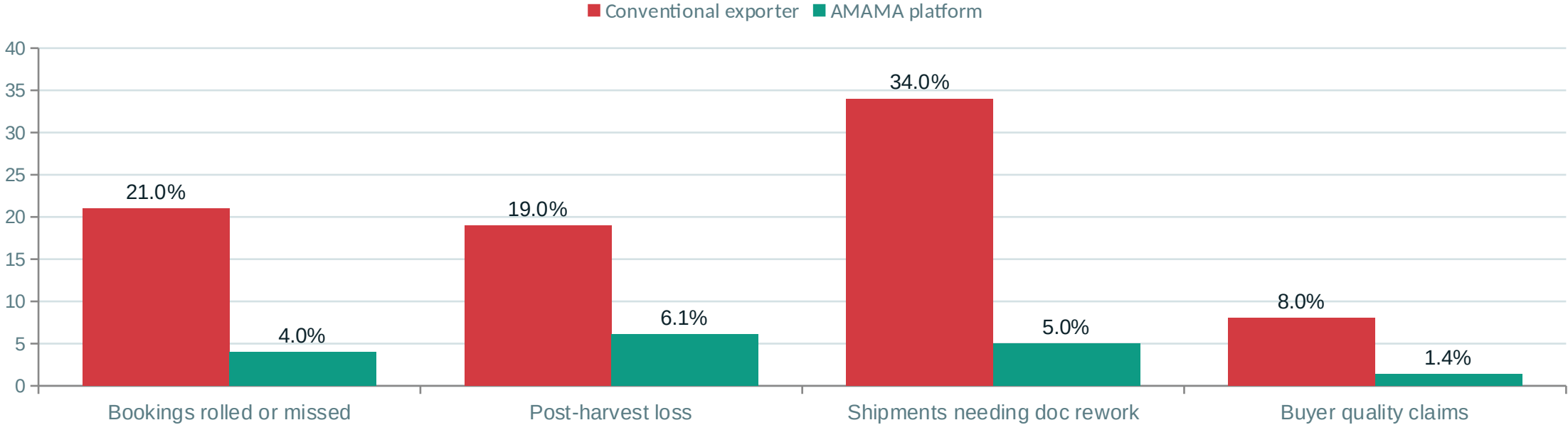
Full-load generator plus rooftop solar

Positioned as common-use export infrastructure: our own exports anchor utilisation, third-party exporters and FPOs fill the balance.

THE CORE OPERATING THESIS

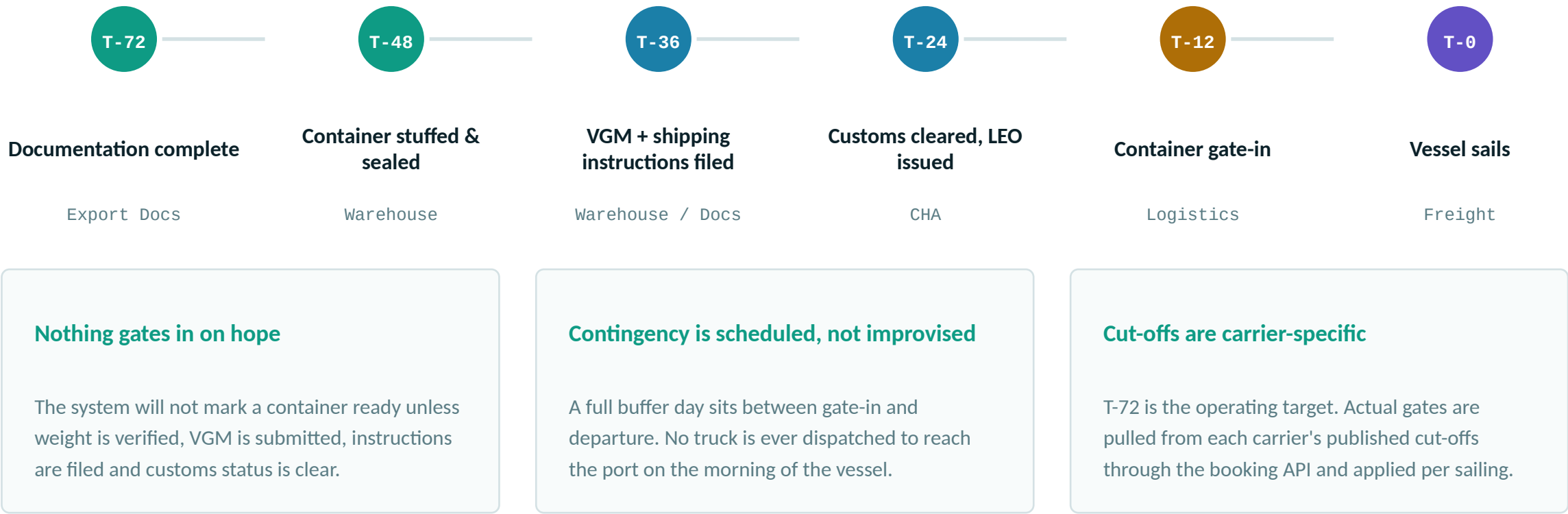
**We do not find a vessel when the cargo is ready.
We secure the vessel before the cargo exists.**

That single inversion is where the on-time rate comes from. Everything downstream — the cut-off engine, the vessel ladder, the risk score — exists to protect a booking that was made before harvest.



The T-72 rule, enforced by the system

Every shipment works backwards from the vessel. A gate that is not cleared on time raises its own alarm, to a named owner rather than to a group inbox.



Never one booking. Always a ladder.

Three to five carriers held open on every lane, scored continuously and switchable in one click.

ROLE	CARRIER / VESSEL	VOYAGE	ETD	TRANSIT	ROUTING	PLUGS	USD/40RH	ROLLOVER	FIT
BOOKED	Maersk · Cape Coral	0426W	+34 h	6 d	Direct	42	\$2,180	11%	84
BACKUP 1	CMA CGM · Dutch Mate	0AB2X	+82 h	8 d	Direct	60	\$2,340	9%	78
BACKUP 2	Hapag-Lloyd · Bremen Express	114W	+106 h	9 d	Jebel Ali T/S	31	\$2,410	7%	71
STANDBY	MSC · Aurora	2609E	+58 h	7 d	Direct	18	\$1,960	24%	52
STANDBY	COSCO · Fortune	091W	+154 h	10 d	Direct	55	\$1,890	19%	49

Fit blends historic rollover rate on the lane, declared reefer plugs, routing, transit days and slack against the T-72 gates.

Backup 1

Alternative sailing, already quoted

Backup 2

Alternative carrier and port pair

Backup 3

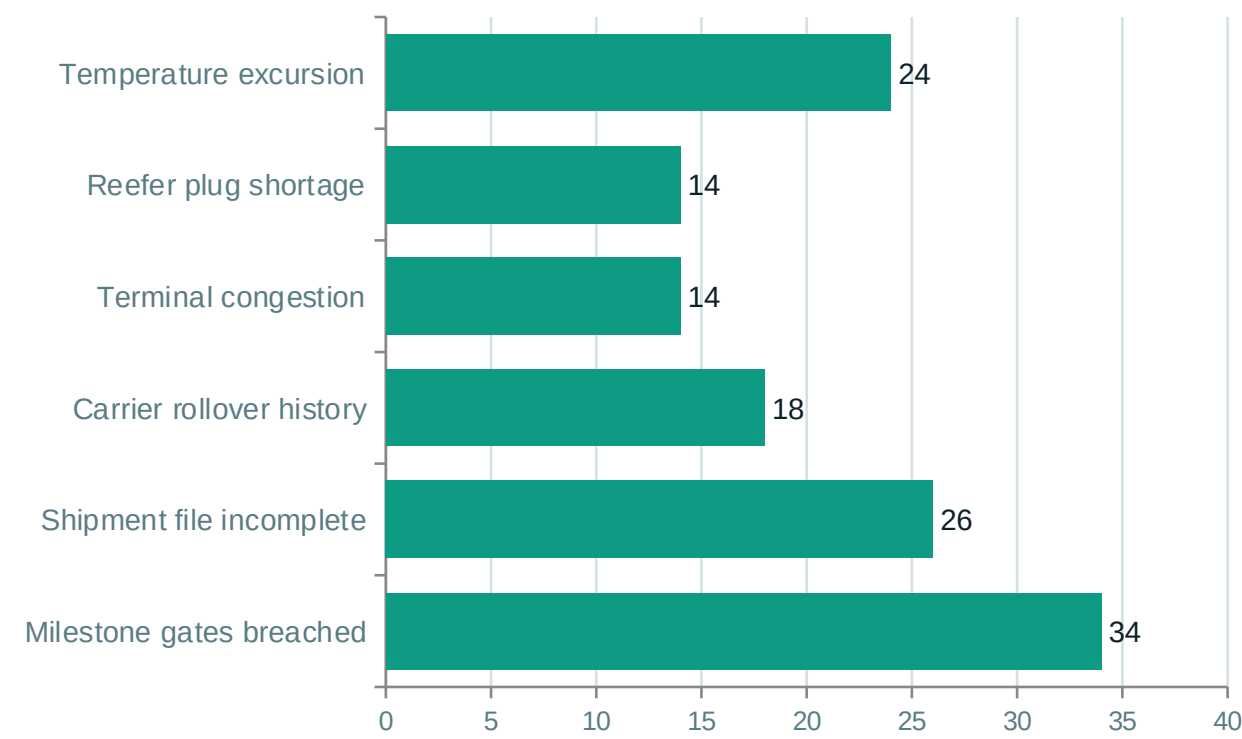
Alternative trucking and CFS route

Backup 4

Emergency cold storage and domestic buyer

Every shipment carries a live score

Recomputed continuously from milestone state, carrier history, terminal congestion and reefer telemetry — and every driver in it resolves to an owner.



0–31 **On track**

Gates clearing with slack. No action.

32–59 **Watch**

Owner notified, backup sailing re-priced, buyer pre-warned.

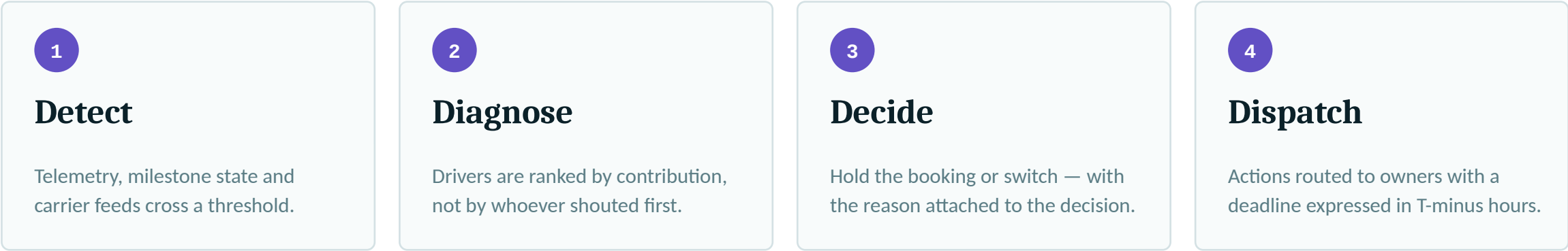
60–100 **Critical**

Exception engine runs, switch recommendation issued, escalation to the duty manager.

A score is only useful if it names a person. Every driver resolves to a role in the responsibility matrix.

When something breaks, the plan writes itself

The engine reads the live shipment state and returns a sequenced action plan, a vessel recommendation and a draft note to the buyer.



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SAMPLE OUTPUT · EXP-DXB-POM-2026-0087

Rollover probability 38% — MSC Aurora is 24% historic on this lane and has 18 plugs against 12 required.

T-30 h Freight · Switch to Maersk Cape Coral 0426W, direct, 42 plugs, one day earlier arrival.

T-26 h Docs · Amend shipping instructions and re-file; carrier SI cut-off moves forward 22 hours.

T-20 h Warehouse · Re-verify VGM against the new booking and re-arm the data logger at 5°C.
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Live in the working build — the control tower calls the engine directly from the shipment screen.

One lot, one identity, from the plant to the plate

The buyer scans a carton and sees the farm. That is the export premium, and it is also the defence when a claim arrives.

AMAMA - POM - MH - 2026 - 000127

brand · commodity · state · season · sequence

Where the record is not optional

For regulated horticultural exports, farm registration, lot creation, laboratory linkage and packhouse declaration flow through APEDA's traceability systems. We generate that record as a by-product of normal operations rather than as a scramble before shipment.

- **FARMER** Sanjay Patil, Dindori, Nashik
- **HARVEST** 10 September, morning window
- **FIELD QC** Grade A+, nine parameters, four photos
- **TRANSIT** Reefer held 5.2–5.6°C, seal unbroken
- **HUB** Pre-cooled to core 5°C in 6 h 40 m

- **STORAGE** CR-01, 5°C / 92% RH, 31 h dwell
- **EXPORT QC** Residue panel cleared, inspection certificate issued
- **CONTAINER** MSKU7841203, PTI passed, logger armed
- **VESSEL** Maersk Cape Coral 0426W ex JNPA
- **DESTINATION** Jebel Ali — Al Maha Fresh FZC

Every controlled document has a name against it

Twenty-eight of them. Responsible does the work, accountable answers for it, and the matrix is enforced by the system rather than stored in a folder.

STAGE	DOCUMENT	RESPONSIBLE	ACCOUNTABLE	SYSTEM OF RECORD	DUE
Farm QC	Grading sheet & lot ID allotment	Field QC Inspector	Head — Quality	Traceability	At pickup
Export QC	Laboratory / residue test report	Compliance Executive	Head — Quality	LIMS integration	T-96 h
Commercial	Commercial invoice & packing list	Export Documentation Exec	Manager — Docs	Docs module	T-72 h
Plant health	Phytosanitary certificate	Compliance Executive	Head — Compliance	PQIS	T-72 h
Freight	Shipping instructions	Export Documentation Exec	Manager — Docs	Carrier API	T-36 h
Terminal	VGM declaration	Warehouse Supervisor	Hub Manager	Weighbridge feed	T-36 h
Customs	Shipping bill & e-SANCHIT upload	CHA / Customs Broker	Manager — Docs	ICEGATE	T-48 h
Customs	Let Export Order	CHA / Customs Broker	Manager — Docs	ICEGATE	T-24 h
Post-shipment	Bill of Lading	Freight Executive	Manager — Freight	Carrier API	T+48 h

Nine of twenty-eight shown. The full matrix is live in the platform, filterable by stage, owner or system, and it gates the shipment: nothing moves to the terminal until the file closes.

Seventeen integrations behind one event contract

Adding a carrier or a terminal is an adapter, not a rebuild. Every adapter normalises into the same event before anything downstream sees it.

FREIGHT

- Carrier eBooking
- Vessel schedules
- Track & trace events

COLD CHAIN

- Reefer telemetry
- Truck GPS & temp logger
- Cold room IoT

COMPLIANCE

- APEDA HortiNet
- Plant Quarantine PQIS

CUSTOMS

- ICEGATE e-SANCHIT
- Shipping bill status

FINANCE

- GST e-Invoice
- E-way bill
- Farmer payout
- Trade finance

INTELLIGENCE

- Risk & exception engine
- Mandi price feed
- Farmer notifications

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{ "type":"container.gate_in", "refs":{"shipment":"EXP-DXB-POM-2026-0087"}, "risk_delta":-14 }
```

Forty-two typed events across six domains. The risk score, the buyer portal, the farmer's WhatsApp message and the control tower all read the same stream — which is why the numbers agree.

Nine dashboards, one source of truth

Each stakeholder sees the slice they can act on, and every action they take moves the same shipment record.

Executive

Throughput, on-time rate, loss, revenue mix

Field & procurement

Forecasts, farm QC, grading, lot creation

Mumbai hub

Dock slots, zone use, cold room telemetry

Freight control tower

Cut-off ladder, risk score, vessel switching

Documentation

File completion, owners, customs status

Buyer portal

Order gates, ETA, QR traceability dossier

Exceptions

Faults with photo evidence, routed to an owner

Responsibility matrix

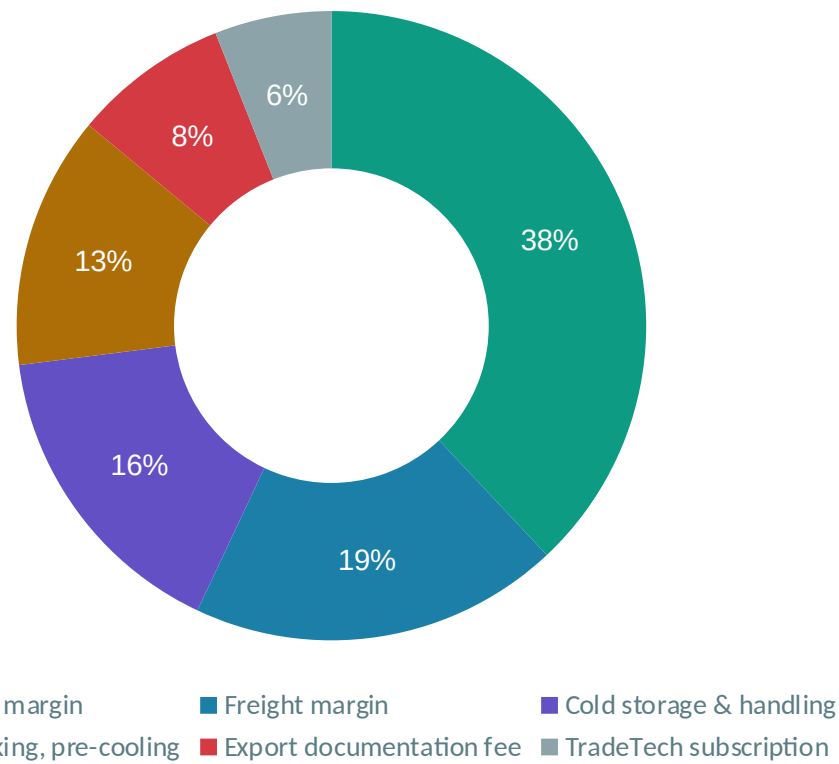
28 documents, filterable, always current

Integration layer

Adapter health, latency, event contract

Ten streams on one asset base

Procurement margin is the obvious one. It is also the one we depend on least.



Third-party utilisation is the point

Charging exporters and FPOs for storage, grading, pre-cooling and packing turns a cost centre into common-use infrastructure with its own P&L.

Freight margin scales without capex

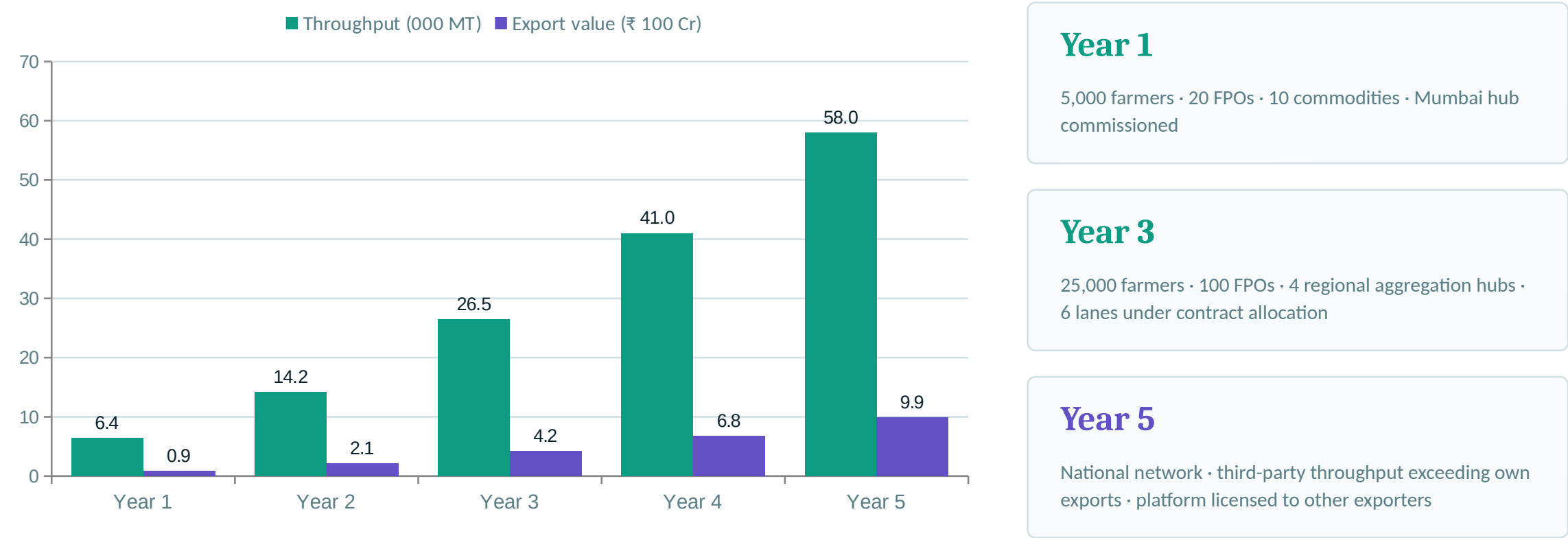
Contracted allocations bought at volume and sold at service — margin that grows with throughput, not with square feet.

Platform revenue is the option value

Once external buyers and sellers transact on the traceability and documentation layer, the software stops being an overhead.

Five years, three phases

Phase one proves the Mumbai hub. Phase two replicates the regional layer. Phase three is a national network on the same spine.



Projections are the working plan for the DPR and are to be restated against verified capex, working-capital and scheme parameters before submission.

The ask is larger than capital, and easier to grant

Positioned as common-use export infrastructure with an anchor tenant, not a private warehouse seeking a subsidy.

- A

Capital support
Cold storage, packhouse, sorting and grading lines, pre-cooling, reefer fleet, QC laboratory.
- B

Land & lease
Access to suitable industrial land inside the Mumbai / JNPA logistics corridor.
- C

Enabling infrastructure
Power, road access, water, drainage, waste handling and connectivity to the site.
- D

Export facilitation
Across APEDA, Customs, Plant Quarantine, laboratories and port authorities.
- E

Farmer mobilisation
Introductions to FPOs and cooperatives in the target production clusters.

- Agriculture Infrastructure Fund**

Recognises warehouses, cold stores, sorting and grading, packhouses and logistics facilities as eligible post-harvest infrastructure.
- APEDA infrastructure assistance**

Support categories covering grading, washing, packaging, palletisation, pre-cooling, cold storage and integrated post-harvest systems.
- MoFPI integrated cold chain scheme**

Grant support for cold chain and value-addition infrastructure, subject to scheme conditions and project ceilings.

No subsidy percentage is quoted until the applicable scheme guidelines, applicant category, project structure and current application window have been verified.

WHAT WE ARE ACTUALLY BUILDING

**Anyone can buy a cold store.
Almost nobody can tell you where
1,240 kg of Grade-A pomegranate is,
right now, to the pallet.**

The building is the entry ticket. The spine that runs through it — lot identity, cut-off discipline, a vessel ladder and an exception engine that names an owner — is the business.

Next: detailed project report, 36 sections — operating model, hub layout, cold chain architecture, technology stack, commodity compliance matrix, capex, working capital, five-year financials and scheme mapping.